

MT 1003 – Calculus and Analytical Geometry**Homework #2****Limits and Continuity****Muhammad Ali and Sara Aziz****Instructions:**

1. All homework files must follow the naming format

Section_RollNumber_HW2

(for example, A.26I1234_HW2). Submissions that do not follow this format will not be evaluated.

2. Write your name and roll number clearly on **every page**.
3. Submit your complete homework as a **single scanned PDF file**.
4. Show all important algebraic steps. An unsupported final answer will not receive full credit.
5. For Section B, include the Python code, generated graphs or output tables, screenshots, and written observations in the same PDF file.
6. In your submitted notebook, add comments beginning with **#** to explain every important line of Python code in your own words.

Failure to comply with these instructions may result in zero marks.

Section A: Analytical Work

Q1. Evaluating Limits by Appropriate Algebraic Methods. Evaluate each limit. Before simplifying, state whether direct substitution gives a determinate value or an indeterminate form. Clearly identify the method used.

(a) $\lim_{x \rightarrow 2} (3x^2 - 5x + 4)$

(b) $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

(c) $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4}$

(d) $\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4}$

(e) $\lim_{x \rightarrow 0} \frac{\sqrt{1 + 2x} - 1}{x}$

- (f) $\lim_{x \rightarrow 0} \frac{\sin 5x}{x}$
- (g) $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$

Q2. One-Sided, Infinite, and Squeezed Limits.

- (a) For $f(x) = \frac{|x|}{x}$, find the left-hand and right-hand limits at $x = 0$. Does the two-sided limit exist? Explain.
- (b) For $g(x) = \frac{x+1}{x-2}$, evaluate $\lim_{x \rightarrow 2^-} g(x)$ and $\lim_{x \rightarrow 2^+} g(x)$. Describe the behavior of the graph near $x = 2$.
- (c) Use the Squeeze Theorem to evaluate $\lim_{x \rightarrow 0} x^2 \sin(1/x)$. Your solution must display an appropriate lower and upper bound.

Q3. Continuity at Joining Points. Consider

$$f(x) = \begin{cases} x^2 + k, & x < 2, \\ 3x - 1, & x \geq 2. \end{cases}$$

- (a) Find $\lim_{x \rightarrow 2^-} f(x)$ and $\lim_{x \rightarrow 2^+} f(x)$ in terms of k .
- (b) Find the value of k for which f is continuous at $x = 2$.
- (c) For your value of k , verify all three conditions for continuity at $x = 2$.
- (d) Sketch the resulting continuous function near $x = 2$, using a clear scale and correctly marked endpoints.

Q4. Classifying Discontinuities. For each function, locate every discontinuity, calculate the relevant one-sided limits, and classify the discontinuity as *removable*, *jump*, or *infinite*.

- (a) $p(x) = \frac{x^2 - 4}{x - 2}$
- (b) $q(x) = \begin{cases} x + 1, & x < 1, \\ x + 3, & x \geq 1, \end{cases}$
- (c) $r(x) = \frac{1}{(x + 2)^2}$
- (d) $s(x) = \frac{x^2 - x - 6}{x^2 - 4x + 3}$

Q5. A Server-Response Model. The normalized response of a server is modeled by

$$R(t) = \frac{t^2 - 25}{t - 5}, \quad t \neq 5,$$

where t is time in seconds.

- (a) Simplify $R(t)$ for $t \neq 5$.
- (b) Find $\lim_{t \rightarrow 5} R(t)$.
- (c) Is R continuous at $t = 5$ as presently defined? State which continuity condition fails.
- (d) Define $R(5)$ so that the extended function becomes continuous.
- (e) Explain in one or two sentences why a graphing program may make this removable discontinuity difficult to see.

Section B: Limits and Continuity with Python

This section requires **no previous knowledge of Python**. Follow the steps below, run the complete programs, and answer the questions by examining the graphs and numerical output. Python provides evidence and visualization; it does not replace the algebraic reasoning required in Section A.

Installing Anaconda and Opening Jupyter Notebook

1. Download the Anaconda Distribution from www.anaconda.com/download and install it using the recommended options. If it is already installed, proceed to the next step.
2. Open **Anaconda Navigator** from the Start menu.
3. Find **Jupyter Notebook** and click **Launch**.
4. In the browser window, open the folder in which you want to save your work.
5. Select **New** → **Python 3 (ipykernel)**.
6. Copy each complete program into a separate notebook cell and press **Shift+Enter** to run it.

Alternative: Using Google Colab

You may use **Google Colab** instead of installing Anaconda and Jupyter Notebook. Google Colab runs Python in a web browser and does not require Python to be installed on your computer. Follow these steps:

1. Open <https://colab.research.google.com> in a web browser and sign in with your Google account.
2. Click **New Notebook**. Colab will create a notebook containing an empty code cell.
3. Rename the notebook using the required homework naming format, for example, `A.26I1234.HW2.ipynb`.
4. Copy the complete Python program for Question 6 into the first code cell.
5. Run the cell with the triangular **Run** button, or press **Shift+Enter**. Wait until the graphs appear below the cell.
6. Click **+ Code** to create a new code cell. Copy the program for Question 7 into that cell and run it in the same way.
7. If Colab displays a message asking whether you want to connect to a runtime, click **Connect** and then run the cell again.
8. Colab normally saves the notebook automatically in Google Drive. You may also select **File** → **Download** → **Download .ipynb** to keep a copy on your computer.

9. Take clear screenshots that include the relevant code and its generated graphs or numerical output. Insert those screenshots and your written observations into the single PDF submitted for this homework.

Q6. Visualizing Three Types of Discontinuity. Run the following complete program. The comments beginning with # explain the purpose of each important line.

```

1  # Import NumPy for creating numerical x-values and piecewise
    calculations.
2  import numpy as np
3
4  # Import pyplot for drawing and displaying the graphs.
5  import matplotlib.pyplot as plt
6
7  # Create x-values on both sides of x = 2 for the removable
    discontinuity.
8  x1_left = np.linspace(-2, 1.95, 400)
9  x1_right = np.linspace(2.05, 6, 400)
10
11 # Evaluate (x^2 - 4)/(x - 2) away from the excluded input x =
    2.
12 y1_left = (x1_left**2 - 4)/(x1_left - 2)
13 y1_right = (x1_right**2 - 4)/(x1_right - 2)
14
15 # Create x-values on the two sides of x = 1 for the jump
    function.
16 x2_left = np.linspace(-3, 0.99, 400)
17 x2_right = np.linspace(1, 5, 400)
18
19 # Evaluate the left and right formulas of the jump function.
20 y2_left = x2_left + 1
21 y2_right = x2_right + 3
22
23 # Create x-values on both sides of x = -2 for the infinite
    discontinuity.
24 x3_left = np.linspace(-6, -2.05, 400)
25 x3_right = np.linspace(-1.95, 2, 400)
26
27 # Evaluate 1/(x + 2)^2 away from x = -2.
28 y3_left = 1/(x3_left + 2)**2
29 y3_right = 1/(x3_right + 2)**2
30
31 # Create one figure containing three graphs in one row.
32 fig, axes = plt.subplots(1, 3, figsize=(16, 4.5))
33
34 # Plot both pieces of the function with a removable
    discontinuity.
35 axes[0].plot(x1_left, y1_left, color="blue", linewidth=2)

```

```

36 axes[0].plot(x1_right, y1_right, color="blue", linewidth=2)
37
38 # Mark the missing point (2, 4) with an open circle.
39 axes[0].scatter([2], [4], facecolors="white", edgecolors="
    blue", s=80)
40
41 # Add a title, axis labels, coordinate axes, and a grid.
42 axes[0].set_title("Removable discontinuity")
43 axes[0].set_xlabel("x")
44 axes[0].set_ylabel("p(x)")
45 axes[0].axhline(0, color="black", linewidth=0.8)
46 axes[0].axvline(0, color="black", linewidth=0.8)
47 axes[0].grid(True, alpha=0.3)
48
49 # Plot the left and right pieces of the jump function.
50 axes[1].plot(x2_left, y2_left, color="purple", linewidth=2)
51 axes[1].plot(x2_right, y2_right, color="purple", linewidth=2)
52
53 # Mark the open left endpoint and filled right endpoint at x
    = 1.
54 axes[1].scatter([1], [2], facecolors="white", edgecolors="
    purple", s=80)
55 axes[1].scatter([1], [4], color="purple", s=45)
56
57 # Add a title, axis labels, coordinate axes, and a grid.
58 axes[1].set_title("Jump discontinuity")
59 axes[1].set_xlabel("x")
60 axes[1].set_ylabel("q(x)")
61 axes[1].axhline(0, color="black", linewidth=0.8)
62 axes[1].axvline(0, color="black", linewidth=0.8)
63 axes[1].grid(True, alpha=0.3)
64
65 # Plot both branches of the function with an infinite
    discontinuity.
66 axes[2].plot(x3_left, y3_left, color="red", linewidth=2)
67 axes[2].plot(x3_right, y3_right, color="red", linewidth=2)
68
69 # Draw the vertical line x = -2 and limit the displayed y-
    values.
70 axes[2].axvline(-2, color="black", linestyle="--", linewidth
    =1)
71 axes[2].set_ylim(0, 20)
72
73 # Add a title, axis labels, coordinate axes, and a grid.
74 axes[2].set_title("Infinite discontinuity")
75 axes[2].set_xlabel("x")
76 axes[2].set_ylabel("r(x)")

```

```

77 axes[2].axhline(0, color="black", linewidth=0.8)
78 axes[2].grid(True, alpha=0.3)
79
80 # Adjust the spacing so the titles and labels do not overlap.
81 plt.tight_layout()
82
83 # Display all three graphs.
84 plt.show()

```

- Paste a screenshot of the three generated graphs into your homework.
- For the first graph, estimate $\lim_{x \rightarrow 2^-} p(x)$ and $\lim_{x \rightarrow 2^+} p(x)$. Why is the discontinuity removable?
- For the second graph, estimate both one-sided limits at $x = 1$. Explain why the two-sided limit does not exist.
- For the third graph, describe the behavior as $x \rightarrow -2^-$ and $x \rightarrow -2^+$. What vertical asymptote is visible?
- State one visual feature that distinguishes each of the three types of discontinuity.
- Explain one limitation of deciding continuity only from a plotted graph.

Q7. Numerically Estimating a Trigonometric Limit. The following program approaches zero from the left and right, prints a table of values of $\sin x/x$, and draws a graph near $x = 0$.

```

1  # Import NumPy for numerical calculations and arrays.
2  import numpy as np
3
4  # Import pyplot for drawing the graph.
5  import matplotlib.pyplot as plt
6
7  # Store selected negative and positive x-values approaching
   zero.
8  x_values = np.array([-0.1, -0.01, -0.001, -0.0001,
9                      0.0001, 0.001, 0.01, 0.1])
10
11 # Calculate sin(x)/x at every selected nonzero input.
12 y_values = np.sin(x_values)/x_values
13
14 # Print a heading for the numerical table.
15 print("          x          sin(x)/x")
16
17 # Print each input and output with ten digits after the
   decimal point.
18 for x, y in zip(x_values, y_values):
19     print(f"{x:12.6f}      {y:14.10f}")
20

```

```

21 # Create many negative x-values close to zero.
22 x_left = np.linspace(-0.5, -0.001, 500)
23
24 # Create many positive x-values close to zero.
25 x_right = np.linspace(0.001, 0.5, 500)
26
27 # Plot sin(x)/x on the left and right of zero.
28 plt.plot(x_left, np.sin(x_left)/x_left, color="darkgreen",
29          linewidth=2)
30
31 # Mark the missing point at (0, 1) with an open circle.
32 plt.scatter([0], [1], facecolors="white", edgecolors="
33             darkgreen", s=90)
34
35 # Add a horizontal reference line at y = 1.
36 plt.axhline(1, color="gray", linestyle="--", linewidth=1)
37
38 # Add a title and label both axes.
39 plt.title("Numerical and Graphical Study of sin(x)/x")
40 plt.xlabel("x")
41 plt.ylabel("sin(x)/x")
42
43 # Restrict the vertical scale to display the limiting
44 # behavior clearly.
45 plt.ylim(0.95, 1.01)
46
47 # Add a grid and display the graph.
48 plt.grid(True, alpha=0.3)
49 plt.show()

```

- Paste a screenshot of the printed table and graph into your homework.
- From the negative inputs, estimate the left-hand limit as $x \rightarrow 0^-$.
- From the positive inputs, estimate the right-hand limit as $x \rightarrow 0^+$.
- Use the two observations to estimate $\lim_{x \rightarrow 0} \sin x/x$.
- Is the displayed formula defined at $x = 0$? Is it continuous there?
- Define a new value at $x = 0$ that makes the extended function continuous, and justify your answer using the continuity conditions.
- Explain why numerical evidence suggests a limit but does not constitute an algebraic proof valid for every sufficiently small nonzero x .